

# BA2: Digital Korea

## Week 10: Topic Modeling (LDA)

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# Today's Agenda

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1. Review & check-in: from feeling to theme
2. What is topic modeling?
3. LDA: the intuition
4. Choosing  $k$  (the number of topics)
5. Orange's Topic Modelling widget under the hood
6. Application: the NIKH Korean history textbook corpus
7. Interpreting topics responsibly
8. Looking ahead

# Review & Check-In

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# From Feeling to Theme

Week 9 asked *how* a text feels; Week 10 asks *what* it is about

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**Last week (sentiment):** a text has a **tone** —positive or negative.

감사 + 희망 + 위기  $\Rightarrow$  positive/negative score

**This week (topics):** a text is about **several things at once**, in different proportions.

조선 왕조 문화  $\Rightarrow$  a *mixture* of themes

*Sentiment:* does the text feel positive or negative?

*Topics:* what themes run through this collection of texts?

# Two Classification Styles, One Week Apart

Both use a bag of words —they ask different questions of it

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## Dictionary sentiment (Week 9)

- *Supervised by a lexicon* —you bring the word lists
- One score per document
- Output: a *number* (positive or negative)

## Topic modeling (Week 10)

- *Unsupervised* —no lexicon, no labels
- Many weights per document
- Output: a *set of themes* and how much each document uses each one

Topic modeling is the first **fully unsupervised** method since clustering (Week 7) —but where clustering gives each document *one* group, LDA gives it a *mixture*.

# What Is Topic Modeling?

# From a Sea of Text to Hidden Structure

The problem: corpora too big to read, too important to ignore

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- The NIKH corpus we will use today covers **67 Korean history textbooks** (1895–2016).
- Even the 11-book demo already runs past a *million characters of Korean prose*.
- You cannot close-read it all. You also cannot reduce it to a single number.

## The topic modeling question

What **themes** run through this collection, **how much** does each document use each theme, and **which words** define each theme?

# Topic Modeling: The Basic Idea

An unsupervised method for discovering hidden thematic structure

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- A **topic** is a group of words that tend to appear together across the corpus.
- Each **document** is represented as a **mixture of topics**.
- The algorithm does not know, in advance, what the topics are —it *discovers* them.

Topic modeling answers three questions at once:

1. What themes appear across the documents?
2. How much of each theme does a given document contain?
3. Which words *define* each theme?



# What a “Topic” Looks Like

Fictional but plausible examples from Korean history textbooks

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A topic is a ranked list of words with weights. We read the top words and *assign* a label ourselves.

| Topic | Top words (Korean)       | Our label     |
|-------|--------------------------|---------------|
| T1    | 고구려, 백제, 신라, 삼국, 왕, 시대   | Ancient Korea |
| T2    | 일제, 독립, 저항, 운동, 항일, 식민   | Colonial era  |
| T3    | 산업화, 경제, 발전, 민주화, 성장, 정부 | Modernization |

The *model* gives us columns 1 and 2; the *researcher* gives column 3.

# LDA: The Intuition

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# Latent Dirichlet Allocation, in Plain Words

The most widely used topic model —and the one Orange runs

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LDA (Blei, Ng & Jordan, 2003) makes two simple assumptions:

1. Each **document** is made up of several **topics**, in some proportion.
2. Each **topic** is made up of **words that tend to occur together**.

*Latent* because the topics are hidden; we only observe words.

*Dirichlet* is a probability distribution that makes the mixtures well-behaved.

*Allocation* because the model allocates each word in each document to one of the topics.

You, the researcher, specify one number: the **number of topics**  $k$ .

# The Generative Story

A thought experiment —how LDA imagines a document is written

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Pretend you are about to write a document. LDA imagines you do this:

1. Pick a **mixture** of topics (“this book will be 70% ancient, 20% colonial, 10% modern”).
2. For each word in the document:
  - pick one *topic* from the mixture,
  - pick one *word* from that topic’s vocabulary,
  - write it down.



LDA runs this story **backwards**: given the words we actually see, it infers the topics and the per-document mixtures.

# Two Things LDA Gives You

Everything else in Orange is a view of these two tables

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## 1. Topic → word weights

*How much does each topic use each word?*

|     | T1   | T2   | T3   |
|-----|------|------|------|
| 고구려 | 0.08 | 0.00 | 0.00 |
| 일제  | 0.00 | 0.09 | 0.01 |
| 산업화 | 0.00 | 0.00 | 0.07 |
| 시대  | 0.04 | 0.02 | 0.02 |

## 2. Document → topic weights

*How much does each document use each topic?*

| Book               | T1   | T2   | T3   |
|--------------------|------|------|------|
| 1940 elementary    | 0.05 | 0.85 | 0.10 |
| 1981 middle school | 0.30 | 0.15 | 0.55 |
| 2002 high school   | 0.35 | 0.10 | 0.55 |

Columns of table 1 *define* the topics; rows of table 2 *describe* the documents.

# Choosing the Number of Topics

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# How Many Topics ( $k$ )?

You pick  $k$  —the algorithm doesn't pick it for you

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- **Too few** ( $k = 2$ ): topics are so broad they aren't useful ("history" vs. "not history").
- **Too many** ( $k = 50$ ): topics fragment, overlap, and fill with noise.
- Try several values and inspect the topics each time.

| Corpus size      | Try $k$ in the range |
|------------------|----------------------|
| 10–50 documents  | 3–6 topics           |
| 50–200 documents | 5–12 topics          |
| 200+ documents   | 10–20 topics         |

For our 11-book NIKH demo we will start with  $k = 4$  or  $k = 5$ .

# The Orange Pipeline

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# The Topic Modelling Widget in Orange

Input a Corpus, pick a method, set  $k$  — read topics out the other side

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## Three methods available

- **LSI** —Latent Semantic Indexing (SVD on the term–document matrix; fast, not probabilistic)
- **HDP** —Hierarchical Dirichlet Process (non-parametric;  $k$  is *inferred*)
- **LDA** —Latent Dirichlet Allocation (our choice this week)

## Main setting for LDA

- Number of topics: the  $k$  you pick
- Results change run to run —LDA uses random initialisation
- Click *Commit* (or enable *Commit Automatically*) to refit

## Why LDA and not LSI or HDP?

LDA produces *probability distributions* that are easy to read as mixtures, and its  $k$  is something you deliberately choose rather than let the algorithm decide. HDP is tempting but harder to interpret for a first pass; LSI is great for search but less natural for labelling themes.

# Inputs and Outputs at a Glance

What the widget expects on the left, what it emits on the right

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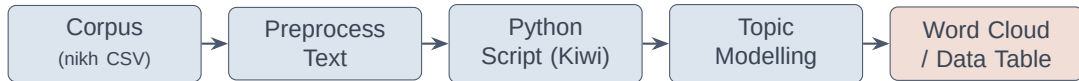
| Port                          | What it carries                                           |
|-------------------------------|-----------------------------------------------------------|
| Input: <i>Corpus</i>          | Your preprocessed documents (tokens, not raw text)        |
| Output: <i>Corpus</i>         | The same corpus with <b>topic weight columns</b> appended |
| Output: <i>Topics</i>         | A table of every topic's top words and weights            |
| Output: <i>All Topics</i>     | The full topic–word distribution ( $\phi$ )               |
| Output: <i>Selected Topic</i> | Just the topic you clicked in the widget                  |

These outputs drive the rest of the workflow: a *Word Cloud* from *Selected Topic*, a *Data Table* from *Corpus*, a *Heatmap* or *Box Plot* grouped by era, and so on.

# A Minimum Orange Workflow

Five widgets from raw CSV to interpretable topics

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- **Corpus.** Load `nikh_clustering_demo.csv`; mark `full_text` as the text feature.
- **Preprocess Text.** Lowercase, tokenise by word, drop stopwords (Korean list provided), filter by document frequency if the corpus is large.
- **Python Script.** Run the Kiwi morphological tokeniser (keep NNG, NNP —*nouns only* for topic modeling).
- **Topic Modelling.** Method = LDA, Number of topics = 4 or 5, click Commit.
- **Word Cloud / Data Table.** Inspect each topic's top words, then open the Corpus output to see the mix per book.

# Nouns Only for Topic Modeling

Why our preprocessing differs from Week 9

Week 9 sentiment kept **nouns, verbs, and adjectives** —feeling lives in verbs and adjectives too. For topic modeling we keep **nouns only**.

| POS tag | Meaning            | Example                    |
|---------|--------------------|----------------------------|
| NNG     | Common noun (일반명사) | 독립 ( <i>independence</i> ) |
| NNP     | Proper noun (고유명사) | 고구려 ( <i>Goguryeo</i> )    |

## Why not verbs?

Topics are about *what a text is about* —and in Korean, *what* is carried by nouns. Verbs like 하다 (*do*) and 되다 (*become*) appear everywhere; they blur topics rather than separate them. Drop them, and the themes sharpen.

# Reading the Output: Top-Words and Topic Weights

Two views of the same LDA fit

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**In the widget itself:** each topic shows as a row with its top words, sized by weight. Click a topic to send its words to a Word Cloud.

**Downstream of the widget:**

- Send the *Corpus* output to a *Data Table* —every row is a book, every new column is a topic weight.
- Send it to a *Box Plot* grouped by era to compare topic use across Colonial / Authoritarian / Democratic textbooks.
- Send the *All Topics* output to a *Heatmap* for a topic–word overview.

## LDavis if you can install it

The LDavis add-on renders topics on an MDS map, with a  $\lambda$  **slider** that blends raw frequency ( $\lambda = 1$ ) with distinctiveness ( $\lambda = 0$ ). Best practice:  $\lambda \approx 0.2$ – $0.35$ .

# Application: The NIKH Corpus

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# The NIKH Corpus

Korean history textbooks from three political eras

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The National Institute of Korean History corpus (국사편찬위원회) collects Korean history textbooks spanning 1895–2016. For today's demo we use the **11-book clustering sample** already on the course website.

| Era                       | Books | Examples                                 |
|---------------------------|-------|------------------------------------------|
| Colonial (1940)           | 3     | 심상소학국사보충아동용 1–2; 교수참고서                   |
| Authoritarian (1973–1987) | 4     | 중학교국사 3 차 (1973); 고등학교국사 4 차 (1981); ... |
| Democratic (1995–2002)    | 4     | 고등학교국사 6 차 (1995); 초등학교사회 6-1 7 차 (2002) |

The corpus is *small in document count* (11) but *large in prose* (tens to hundreds of thousands of characters per book). That is exactly the shape of corpus LDA was designed for.

# Research Questions for the Demo

What we will ask of the NIKH corpus in class

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1. What themes run through Korean history textbooks as a group?
2. Do Colonial-era textbooks emphasise different topics than post-1987 Democratic-era ones?
3. Is there a topic that is more prominent in elementary versus high-school textbooks?
4. Are some topics near-universal (every textbook uses them) and others era-specific?

## A research choice — not a given

The era labels (Colonial / Authoritarian / Democratic) are an *editorial* split from the corpus metadata. Different historians might draw the lines differently — the periodisation is part of your analysis, not a fact handed to you by the algorithm.



# What to Expect in the Demo Output

Plausible topics for an LDA with  $k = 4$  on this corpus

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| Topic | Likely top words        | Our working label           |
|-------|-------------------------|-----------------------------|
| T1    | 고구려, 백제, 신라, 삼국, 왕조, 시대 | Ancient & pre-modern states |
| T2    | 조선, 왕, 시대, 성리학, 문화      | Joseon dynasty              |
| T3    | 일제, 독립, 운동, 저항, 식민, 항일  | Colonial / resistance       |
| T4    | 민주, 경제, 산업화, 발전, 정부, 국민 | Modern Korea                |

What we *expect* to see in the document–topic table:

- 1940 textbooks skewed toward T1–T2 (the Japanese colonial authorities taught a “pre-modern” Korea).
- 1980s textbooks mixing T3 strongly (the colonial period as a topic).
- 2002 textbooks spreading weight across T1–T4 (broadest coverage).

# **Interpreting Topics Responsibly**

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# Strengths of LDA

What the method is good for

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- **Exploratory.** Reveals structure in corpora too large to close-read.
- **Mixture-aware.** A textbook can be 60% colonial-era and 40% modernisation — clustering could not say that.
- **Comparative.** Topic weights are numbers: you can group by era, decade, author, level.
- **Portable.** The same widget works on tweets, speeches, newspaper articles, or textbook chapters.

# Limitations of LDA

What the method will *not* do for you

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- **Topics are statistical, not semantic.** They are co-occurrence patterns; *you* give them names.
- **Sensitive to preprocessing.** Change stopwords, POS filter, or document length — topics change.
- **Sensitive to  $k$ .** Re-running with  $k = 3$  vs  $k = 7$  gives different themes.
- **Non-deterministic.** Two runs with the same  $k$  can produce different topic orderings (and sometimes different topics).
- **Junk topics happen.** Expect one or two topics to be incoherent at any given  $k$ .

## A useful reminder

Grimmer, Roberts & Stewart: *“All quantitative models of language are wrong, but some are useful.”* LDA is a **reading aid**, not an oracle.

# Best Practices for Interpretation

The checklist we will use in the demo

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1. Run LDA twice with the same  $k$ . Are the topics stable? If not, raise or lower  $k$ .
2. For each topic, read the **top 10–15 words**. Draft a label that covers most of them.
3. Open the *Corpus* output and find the 2–3 books with the **highest weight** for each topic. Does the book title fit your label?
4. Look for **one “junk” topic** and be honest about it.
5. Compare topic weights across era — a *Box Plot* grouped by era is usually enough.

# Looking Ahead

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# For the Lab

Interactive, workflow, short write-up

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**Explore.** Work through the NIKH clustering interactive on the course website — note which books cluster together and hold that in your head while you run LDA.

**Build.** The 5-widget Orange pipeline on `nikh_clustering_demo.csv`: Corpus → Preprocess → Python Script (Kiwi, nouns only) → Topic Modelling (LDA,  $k = 4$ ) → Word Cloud.

**Write.** 3–4 sentences per topic:

- your label for the topic;
- the top 5 words that earned that label;
- which era(s) use it most, read off the *Corpus* output.

# Where This Leaves Us

A map of the semester so far

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| Week      | Method                             | What it gives you                   |
|-----------|------------------------------------|-------------------------------------|
| 4–5       | BoW / TF–IDF                       | descriptive word frequencies        |
| 7         | Clustering (K-means, hierarchical) | one group per document              |
| 8         | Word embeddings                    | meaning as vectors                  |
| 9         | Dictionary sentiment               | a tone score per document           |
| <b>10</b> | <b>LDA topic modeling</b>          | <b>a theme mixture per document</b> |

**Recommended reading.** Grimmer, Roberts & Stewart — Chapter 13: *Topic Models*.

**Next week.** Week 11: Final review and assessment. Bring questions about the Research Methods Project (Week 12).